

设函数 $f(x)$ 在点 x_0 的某邻域内能展成幂级数, 即有

$$f(x) = \sum_{n=0}^{\infty} a_n (x-x_0)^n, \quad x \in N_{\Delta}(x_0)$$

$f(x)$ 在 $N_{\Delta}(x_0)$ 内具有任意阶导数

$$f^{(n)}(x) = n! a_n + (n+1)! a_{n+1} (x-x_0) + \dots + \frac{(n+2)!}{2!} a_{n+2} (x-x_0)^2 + \dots$$

$$\Rightarrow f^{(n)}(x_0) = n! \cdot a_n$$

$$a_n = \frac{1}{n!} f^{(n)}(x_0) \quad n=0, 1, 2, \dots$$

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(x_0) (x-x_0)^n \quad x \in N_{\Delta}(x_0)$$

求 $\cos x$ 的麦克劳林展开式

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \quad x \in (-\infty, +\infty)$$

$$\text{求} \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \quad x \in (-\infty, +\infty)$$

$$\cos x^{2026} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{4052k}$$

$$f(x) = \ln \frac{1+x}{1-x}$$

$$= \ln(1+x) - \ln(1-x)$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n - \sum_{n=1}^{\infty} \left(-\frac{1}{n}\right) x^n = \sum_{n=1}^{\infty} \frac{2}{2n+1} x^{2n+1} \quad x \in (-1, 1)$$

$\sin x$ 在 $x = \frac{\pi}{4}$ 的泰勒展开式

设 $t = x - \frac{\pi}{4}$, 在 $t=0$ 的幂级数

$$\sin x = \sin\left(t + \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} (\sin t + \cos t)$$

$$\sin t = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n+1}}{(2n+1)!} \quad t \in (-\infty, +\infty)$$

$$\cos t = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{(2n)!} \quad t \in (-\infty, +\infty)$$

$$\Rightarrow \sin x = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{\sqrt{2}}{2} \left(\frac{(x - \frac{\pi}{4})^{2n}}{(2n)!} + \frac{(x - \frac{\pi}{4})^{2n+1}}{(2n+1)!} \right) \quad x \in (-\infty, +\infty)$$

$f(x) = \frac{1}{x^2 + 4x + 3}$ 在 $x=1$ 的泰勒展开式

$$\text{令 } t = x - 1$$

$$f(x) = \frac{1}{(t+2)(t+4)} = \frac{1}{2} \left(\frac{1}{t+2} - \frac{1}{t+4} \right) = \frac{1}{2} \left(\frac{1}{2} \cdot \frac{1}{1+\frac{t}{2}} - \frac{1}{4} \cdot \frac{1}{1+\frac{t}{4}} \right)$$

$$= \frac{1}{2} \left[\frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{t}{2}\right)^n - \frac{1}{4} \sum_{n=0}^{\infty} \left(-\frac{t}{4}\right)^n \right] \quad |t| < 2$$

$$= \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{2^{n+2}} - \frac{1}{2 \cdot 4^{n+1}} \right) (x-1)^n \quad x \in (-1, 3)$$

计算 $\ln(1.2)$ 的近似值, 误差 $\leq 10^{-4}$

$$\ln(1+x)$$
$$\text{求导} = \frac{1}{1+x} = \sum_{n=0}^{\infty} x^n$$

积分

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n \quad x \in (-1, 1]$$

$$\ln(1.2) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (0.2)^n \quad \text{交错级数}$$

$$|R_n| = \left| \sum_{k=n+1}^{\infty} \frac{(-1)^{k-1}}{k} \cdot (0.2)^k \right| \leq \frac{0.2^{n+1}}{n+1} \leq \frac{1}{10^4}$$

$n=4$ 时满足

$$\ln(1.2) \approx 0.2 - \frac{1}{2}(0.2)^2 + \frac{1}{3}(0.2)^3 - \frac{1}{4}(0.2)^4$$

计算 e 的近似值, 精确到 10^{-6}

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n \quad x \in (-\infty, +\infty)$$

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

$$|R_n| = \sum_{k=n+1}^{\infty} \frac{1}{k!} \leq \frac{1}{n!} \left(\frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} + \dots \right)$$

$$< \frac{1}{n!} \cdot \left(\frac{1}{n+1} + \frac{1}{n+1} - \frac{1}{n+2} + \frac{1}{n+2} - \dots \right)$$

$$< \frac{2}{(n+1)!}$$

计算定积分 $\int_0^{\frac{1}{2}} e^{-x^2} dx$ 的近似值, 精确到 10^{-4}

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \dots + (-1)^n \frac{x^{2n}}{n!} + \dots$$

$$\int_0^{\frac{1}{2}} e^{-x^2} dx = \int_0^{\frac{1}{2}} \left(1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \dots + (-1)^n \frac{x^{2n}}{n!} \right) dx$$

$$= \frac{1}{2} \left(1 - \frac{1}{2^2 \cdot 3} + \frac{1}{2^4 \cdot 5 \cdot 2!} - \dots + (-1)^n \frac{1}{2^{2n} (2n+1) n!} + \dots \right)$$

交错级数: $|R_n| \leq \frac{1}{2^{2n+1} (2n+1) \cdot n!}$

$n=4$ 满足.

$$\textcircled{1} e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad x \in (-\infty, +\infty)$$

$$\textcircled{2} \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad x \in (-1, 1)$$

$$\textcircled{3} \ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} \quad x \in (-1, 1]$$

$$\textcircled{4} \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \quad x \in (-\infty, +\infty)$$

$$\textcircled{5} (1+x)^\alpha = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{n!} x^n \quad x \in (-1, 1) \quad \alpha \notin \mathbb{N}$$

$$\textcircled{6} \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \quad x \in (-\infty, +\infty)$$

$$f(x) = \sin x$$

$$f^n(x) = \sin(x + \frac{n\pi}{2}) \quad n=1,2,\dots$$

$$f^n(0) = \sin(\frac{n\pi}{2}) = \begin{cases} 0 & n=2k \\ (-1)^k & n=2k+1 \end{cases} \quad k=0,1,2,\dots$$

$$\Rightarrow \sin x \text{ 的麦克劳林级数为 } x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^k \frac{x^{2k+1}}{(2k+1)!} + \dots$$

收敛半径 $R=+\infty$

$$\text{余项 } R_n \text{ 满足 } |R_n(x)| = \left| \frac{\sin(\theta x + \frac{n-1}{2}\pi)}{(n+1)!} x^{n+1} \right| \leq \frac{|x|^{n+1}}{(n+1)!} \rightarrow 0 \quad (n \rightarrow \infty)$$

$$\Rightarrow \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \quad x \in (-\infty, +\infty)$$

$$f(x) = (1+x)^\alpha$$

$$f^n(x) = \alpha(\alpha-1)\dots(\alpha-n+1)(1+x)^{\alpha-n} \quad n=1,2,\dots$$

$$f(0)=1 \quad f^n(0) = \alpha(\alpha-1)\dots(\alpha-n+1), \quad n=1,2,\dots$$

$$\Rightarrow (1+x)^\alpha \text{ 的麦克劳林级数为 } 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!} x^n \quad x \in (-1,1)$$